CHE 2106

SPECTROSCOPIC METHODS IN ORGANIC CHEMISTRY

Chemical Identification

- Comparison of Physical Properties
  - Boiling Point
  - Melting Point
  - Density
  - Optical rotation
  - Appearance
  - Odor
- Chemical Test
  - Elemental Analysis
    - Burn the compound and measure the amounts of CO_2, H_2O and other components that are produced to determine the empirical formula.
    - Used today as a test of purity of compounds that have already been identified.

Spectroscopy

Spectroscopy is a general term referring to the interactions of various types of electromagnetic radiation with matter.

Exactly how the radiation interacts with matter is directly dependent on the energy of the radiation

Structural Features we can address Spectroscopically

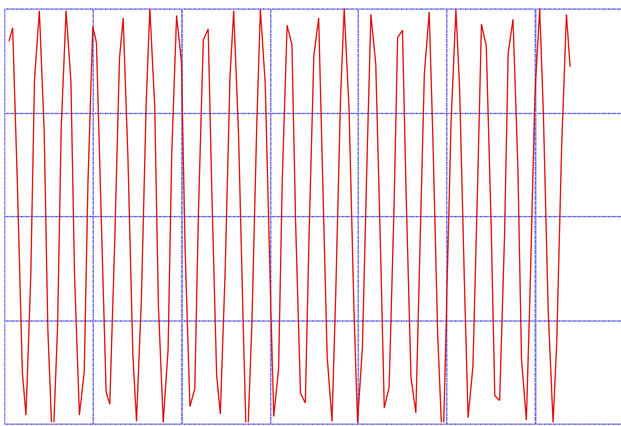
- Molecular weight
- Chemical Formula
- Functional groups
- Skeletal Connectivity, structural isomers
- Spatial-geometric arrangements, stereoisomerism, symmetry
- Presence and location of chromophores
- Chirality issues
- Some of these are more central than others. Sometimes we can stop when the answer is fit to purpose.

WHAT IS SPECTROSCOPY?

- Atoms and molecules interact with electromagnetic radiation (EMR) in a wide variety of ways.
- Atoms and molecules may absorb and/or emit EMR.
- Absorption of EMR stimulates different types of motion in atoms and/or molecules.
- The patterns of absorption (wavelengths absorbed and to what extent) and/or emission (wavelengths emitted and their respective intensities) are called '***spectra***'.
- The field of **spectroscopy** is concerned with the interpretation of ***spectra*** in terms of atomic and molecular structure (and environment).

The Electromagnetic Spectrum

- Light comes in different colors
- No matter what part of range, there are some features in common, that you should know.



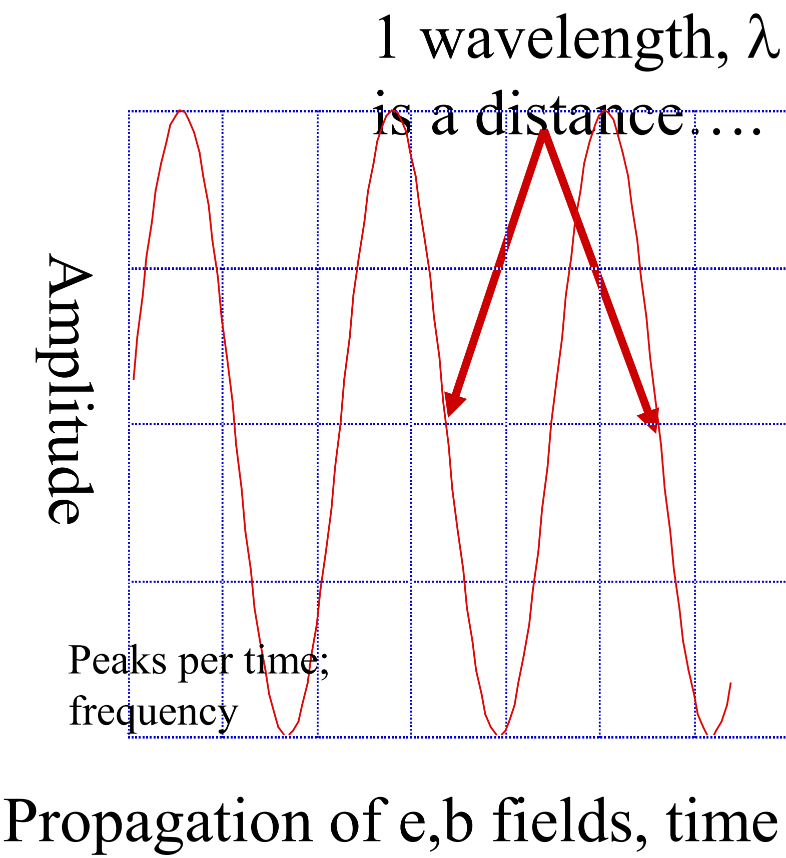
Photon $\Delta E = h\nu$

ν is frequency, Hz, 1/sec

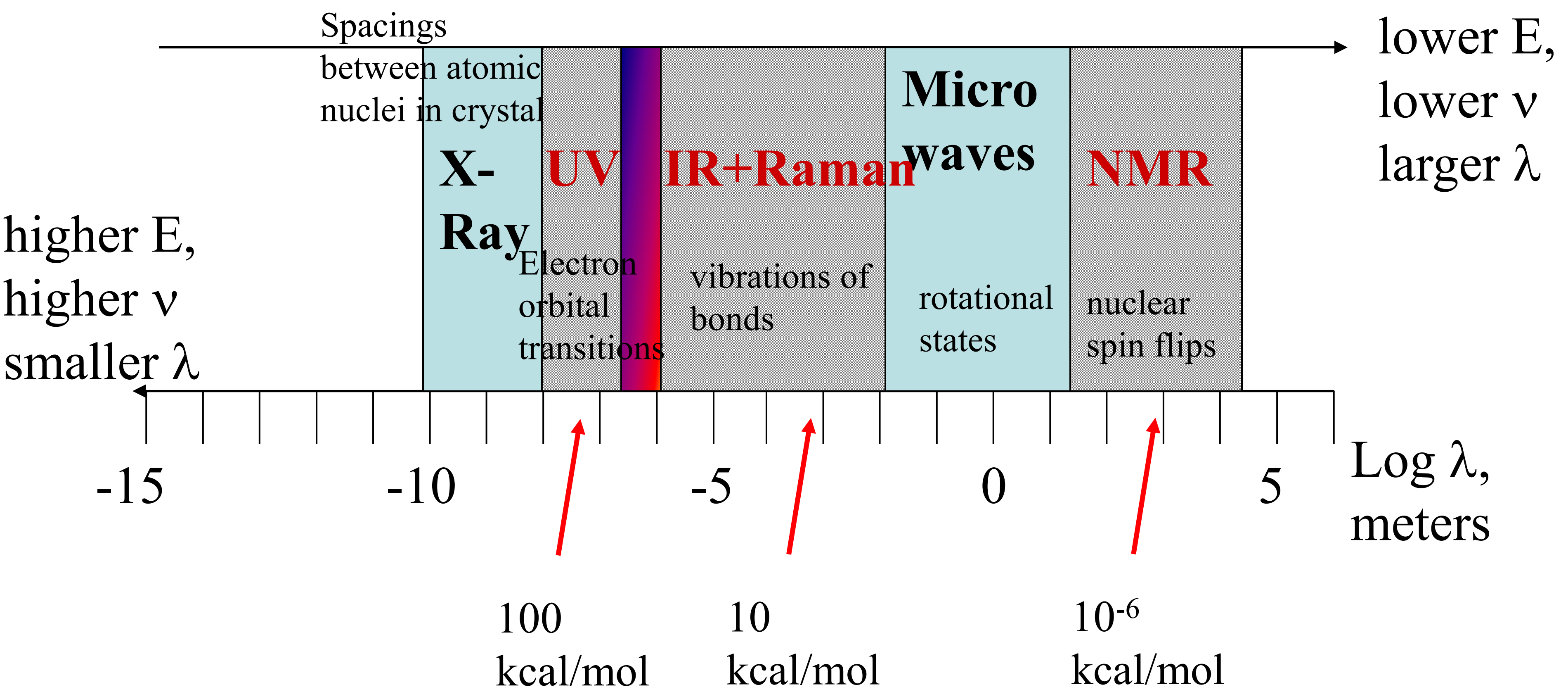
$\nu = c/\lambda$; |$c = 3\text{x}10^{10}$ cm/sec

h = Planck's constant = $6.624 \text{ x}10^{-34}$ J•sec

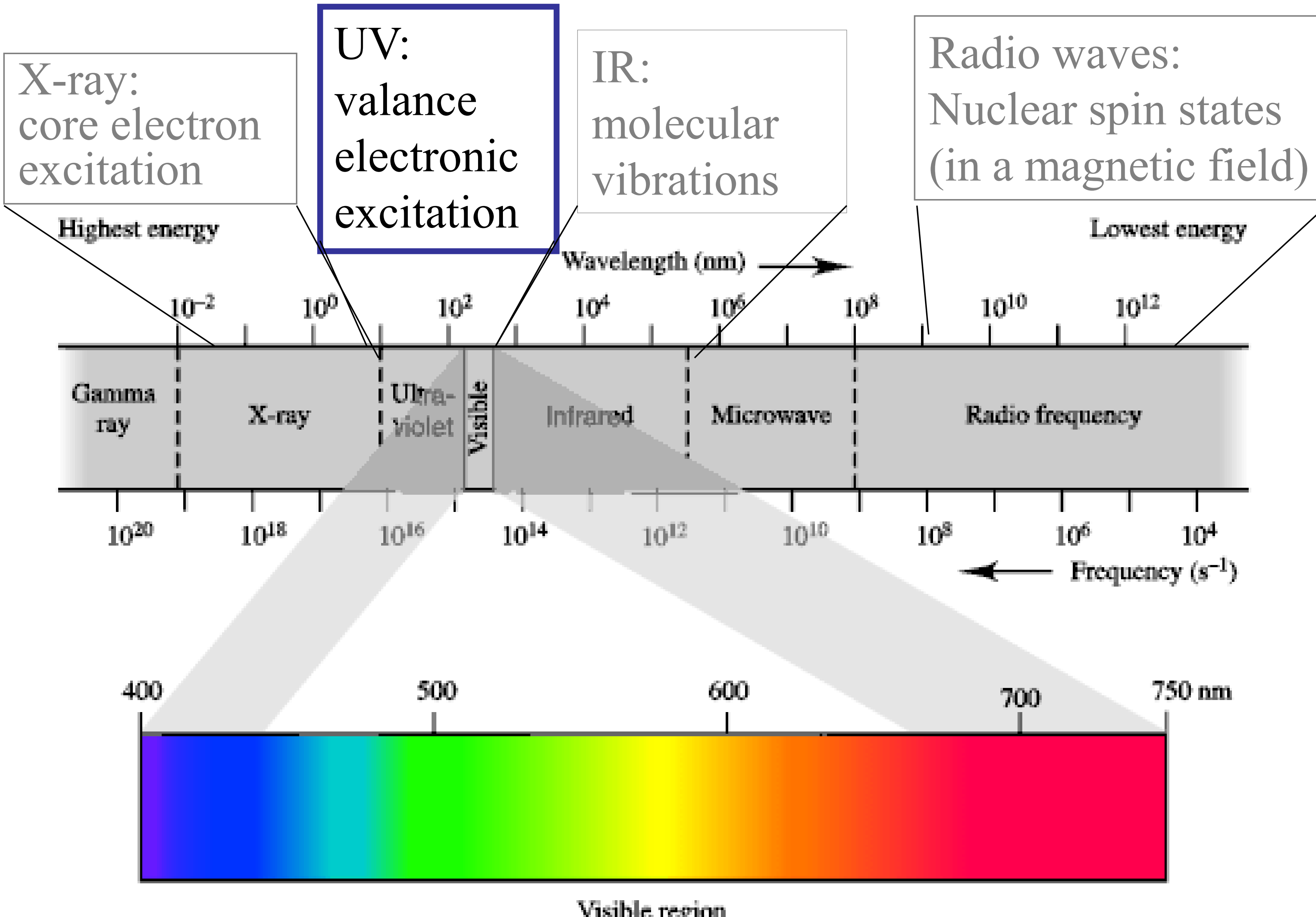
$\nu/c = 1/\lambda$ = wavenumber, cm^{-1}



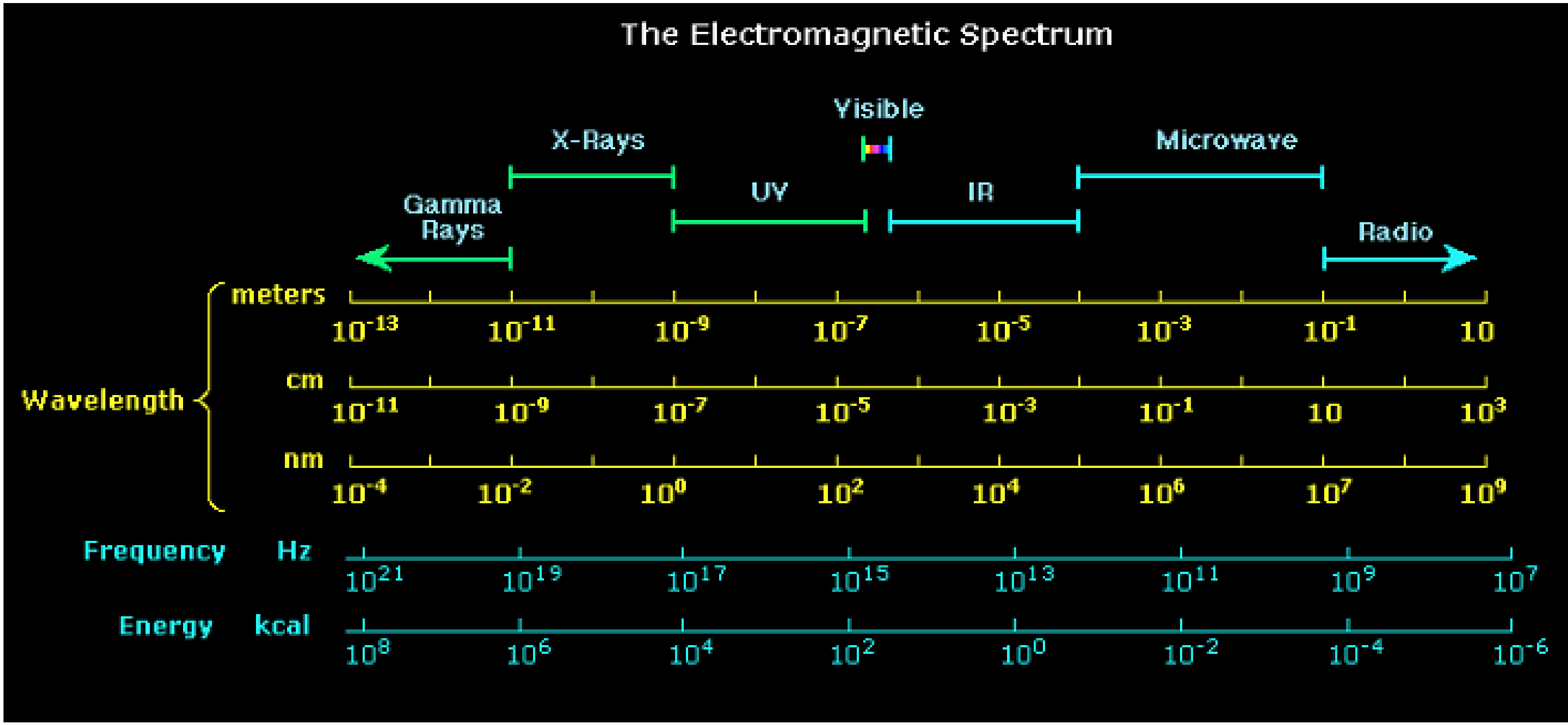
Chemical Properties as related to the different colors of light



Electronic Excitation by UV/Vis Spectroscopy :



THE ELECTROMAGNETIC SPECTRUM



Important: As the wavelength gets shorter, the energy of the radiation increases.

Spectroscopic Techniques

- **NMR** – Nuclear Magnetic Resonance Spectroscopy looks about atoms by means of their nuclei. Connectivity pathways, spatial arrangements of atoms. NMR measures the absorption of radio waves by C and H in a magnetic field. Different kinds of C and H absorb energy of different wavelengths.
- **Mass Spectrometry**--measures molecular weight, most fundamentally useful for unknowns. Controlable fragmentation can distinguish among rival possibilities
- **IR and Raman Spectroscopy**--Vibrations characteristic of bonds, particularly for functional group identification. Excellent “fingerprint”. IR Spectroscopy measures the absorption of infrared (heat) radiation by organic compounds. Different functional groups (C=O, -OH) absorb energy of different wavelengths.
- **UV-/ Visible Spectroscopy**-reports on conjugation and multiple bonds. Provides entre’ to chiroptical probes to assymetric configurations

The higher energy **ultraviolet and visible wavelengths** affect the energy levels of the outer electrons.

Infrared radiation is absorbed by matter resulting in rotation and/or vibration of molecules.

Radio waves are used in nuclear magnetic Resonance and affect the spin of nuclei in a magnetic field.

Spectroscopic Techniques and Chemistry they Probe		
UV-vis	UV-vis region	bonding electrons
Atomic Absorption	UV-vis region	atomic transitions (val. e-)
FT-IR	IR/Microwave	vibrations, rotations
Raman	IR/UV	vibrations
FT-NMR	Radio waves	nuclear spin states
X-Ray Spectroscopy	X-rays	inner electrons, elemental
X-ray Crystallography	X-rays	3-D structure

Spectroscopic Techniques and Common Uses		
UV-vis	UV-vis region	Quantitative analysis/Beer's Law
Atomic Absorption	UV-vis region	Quantitative analysis Beer's Law
FT-IR	IR/Microwave	Functional Group Analysis
Raman	IR/UV	Functional Group Analysis/quant
FT-NMR	Radio waves	Structure determination
X-Ray Spectroscopy	X-rays	Elemental Analysis
X-ray Crystallography	X-rays	3-D structure Anaylysis

Different Spectroscopies

- UV-vis – electronic states of valence e/d-orbital transitions for solvated transition metals
- Fluorescence – emission of UV/vis by certain molecules
- FT-IR – vibrational transitions of molecules
- FT-NMR – nuclear spin transitions
- X-Ray Spectroscopy – electronic transitions of core electrons

UV-Vis Spectroscopy

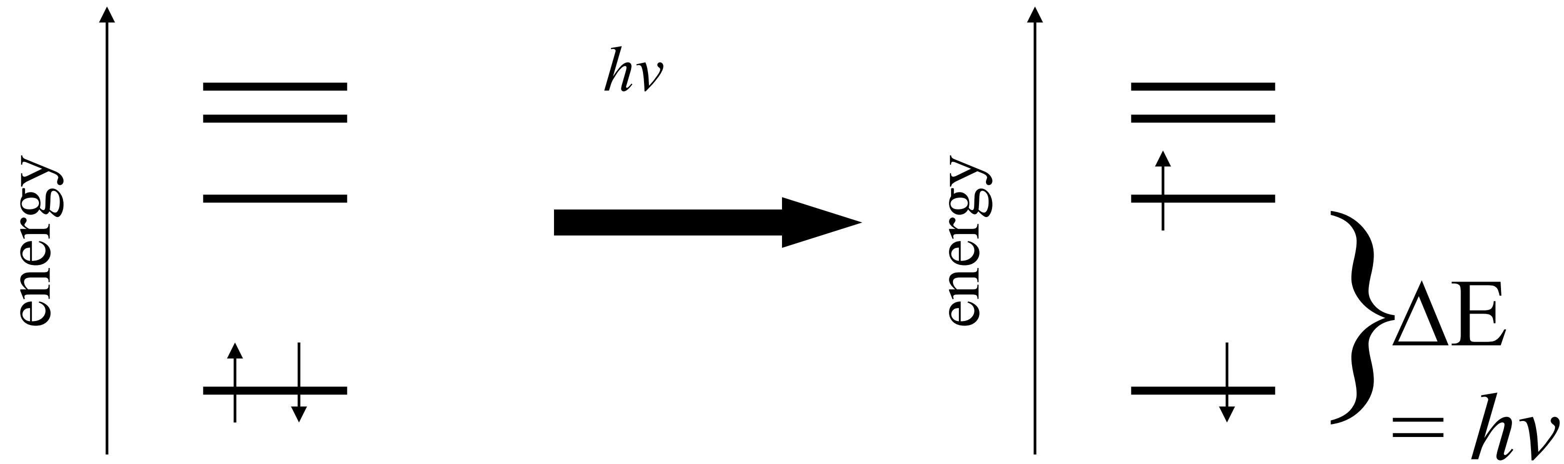
- UV- organic molecules
  - Outer electron bonding transitions
  - conjugation
- Visible – metal/ligands in solution
  - d-orbital transitions
- Instrumentation

Characteristics of UV-Vis spectra of Organic Molecules

- Absorb mostly in UV unless highly conjugated
- Spectra are broad, usually to broad for qualitative identification purposes
- Excellent for quantitative Beer's Law-type analyses
- The most common detector for an HPLC High Performance Liquid Chromatography. "Chromatography" is a technique for separation,

Molecules have quantized energy levels:

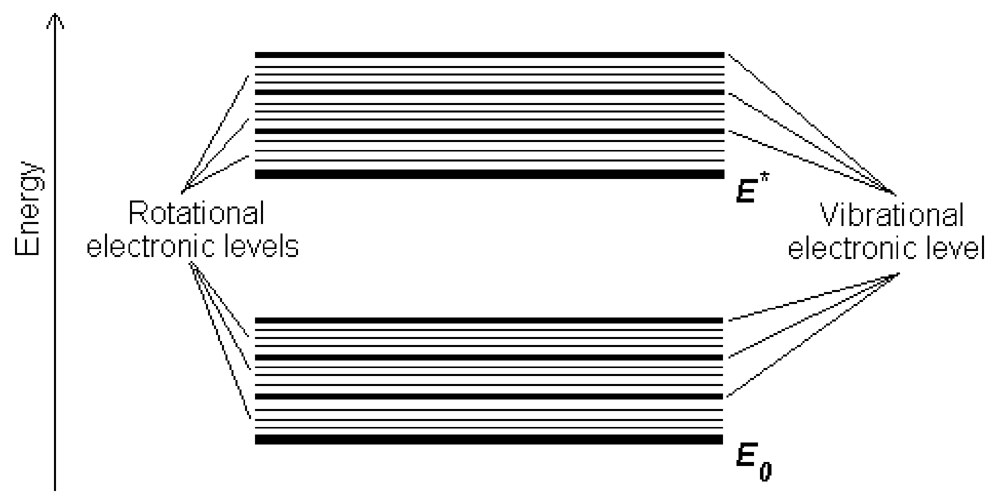
ex. electronic energy levels.

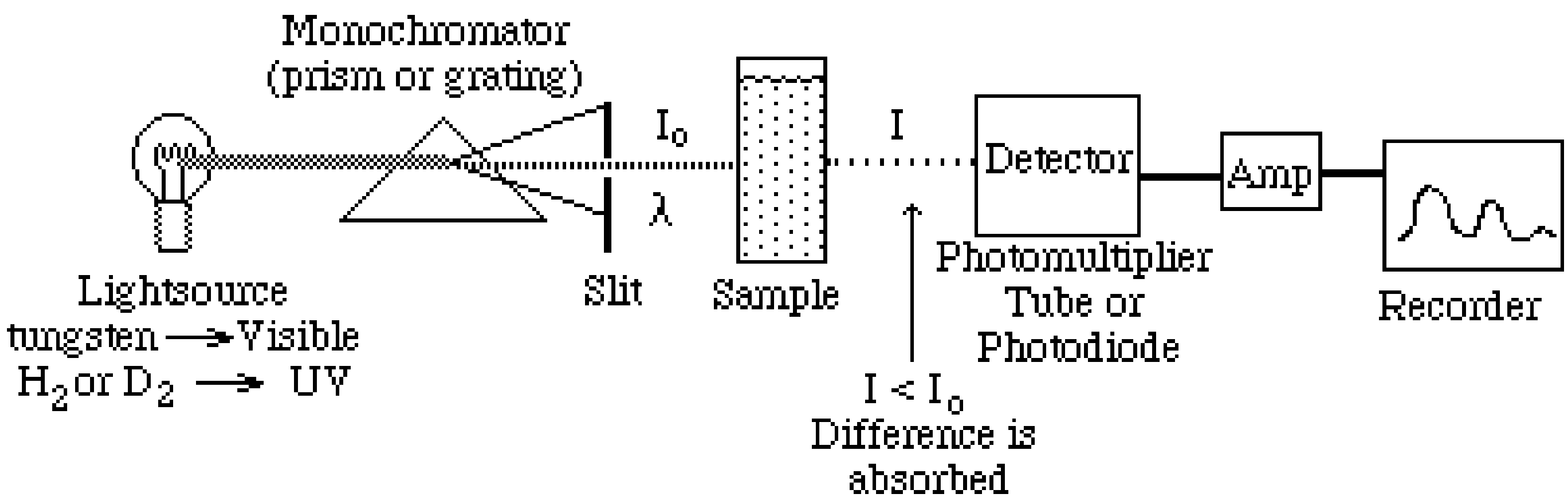


Q: Where do these quantized energy levels come from?

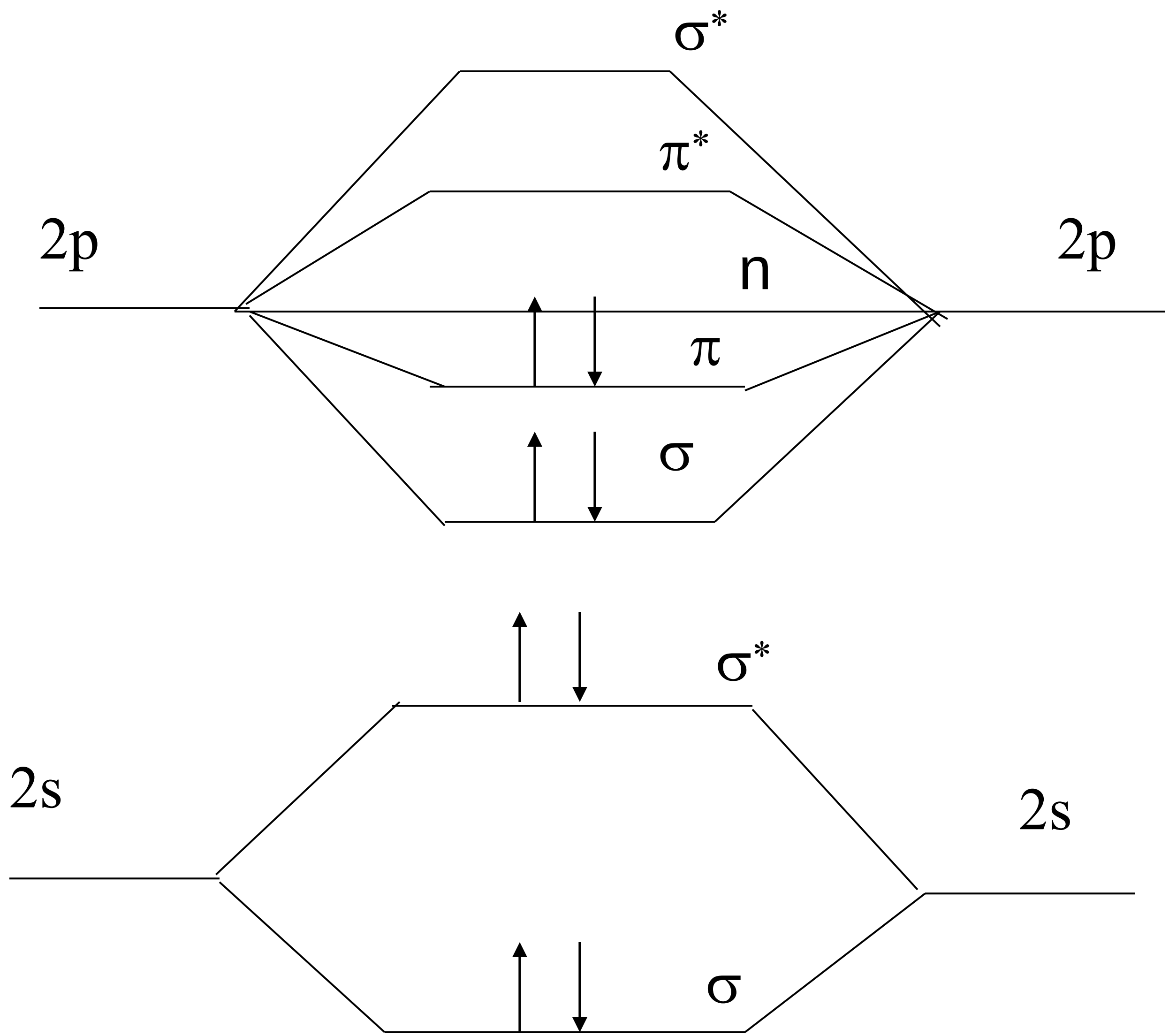
A: The electronic configurations associated with bonding.

Each electronic energy level (configuration) has associated with it the many vibrational energy levels we examined with IR.

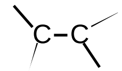


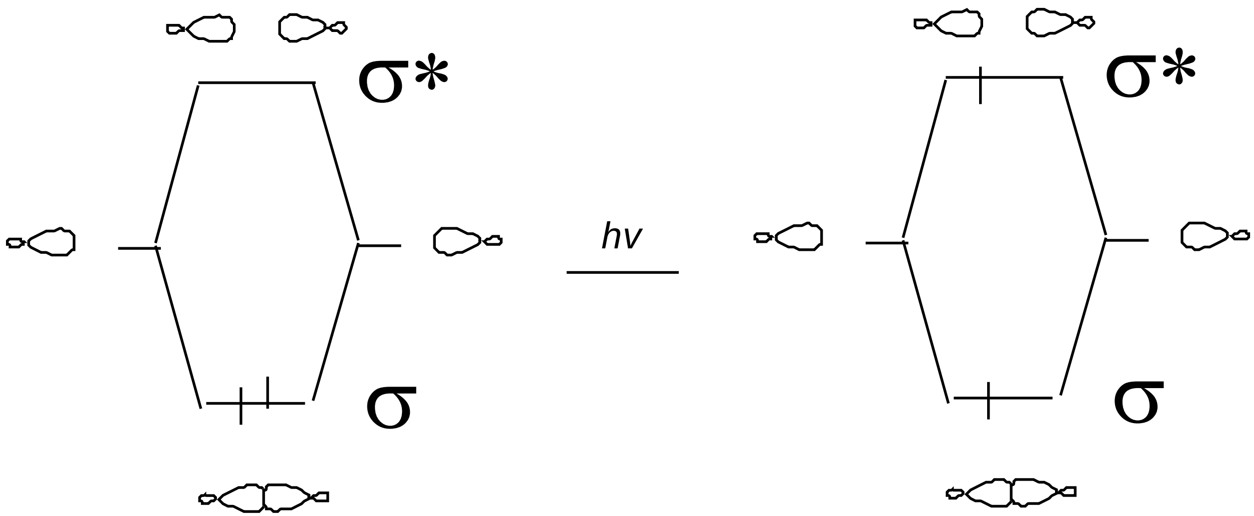

Monochromator
(prism or grating)
I_o
λ
I
Detector
Amp
Lightsource
tungsten → Visible
H_2 or D_2 → UV
Slit
Sample
Photomultiplier
Tube or
Photodiode
Recorder
$I < I_o$
Difference is
absorbed

Molecular Orbital Theory


σ*
π*
2p
n
2p
π
σ
σ*
2s
2s
σ

Ethane



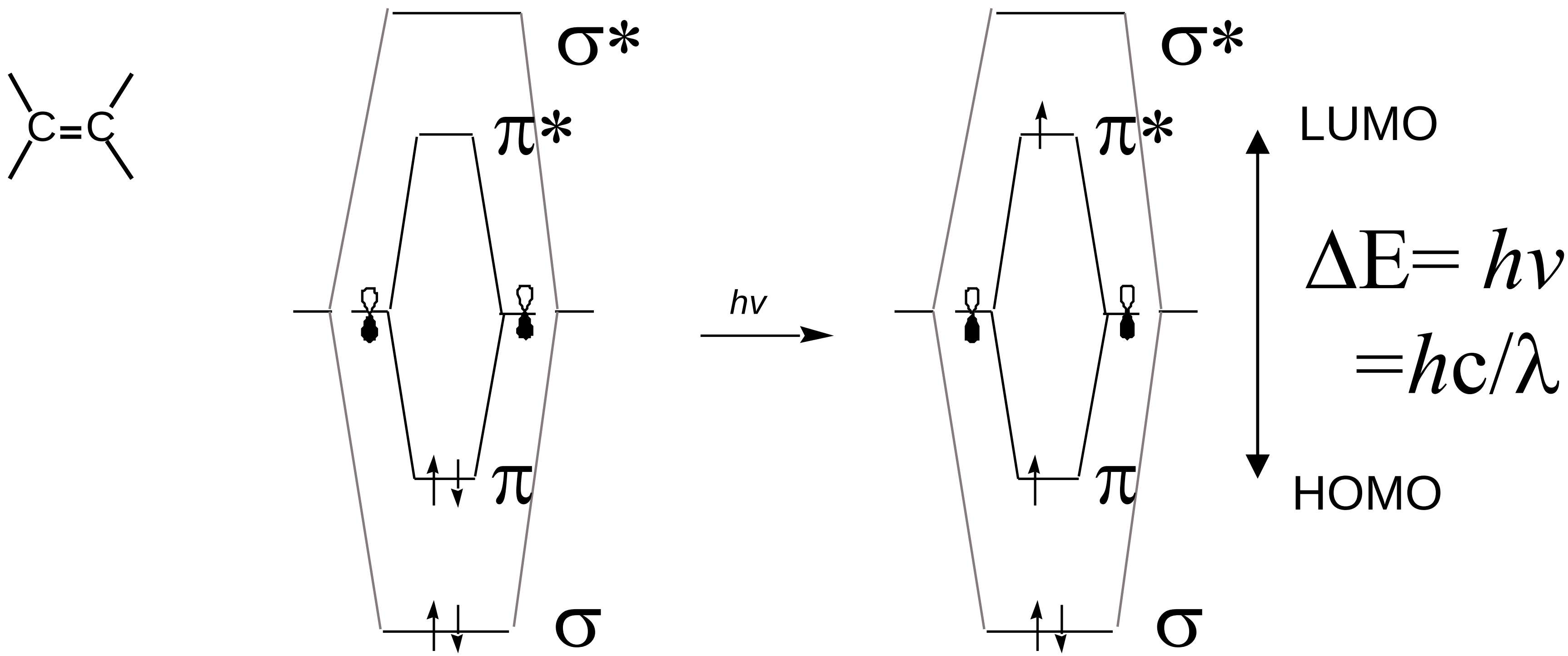


$$\sigma \longrightarrow \sigma^*$$

λ_{max} = 135 nm   (a high energy transition)

Absorptions having λ_{max} < 200 nm are difficult to observe because everything (including quartz glass and air) absorbs in this spectral region.

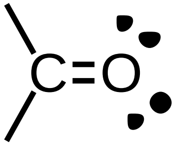
Preferred transition is between
Highest Occupied Molecular Orbital (HOMO) and Lowest Unoccupied Molecular Orbital (LUMO).

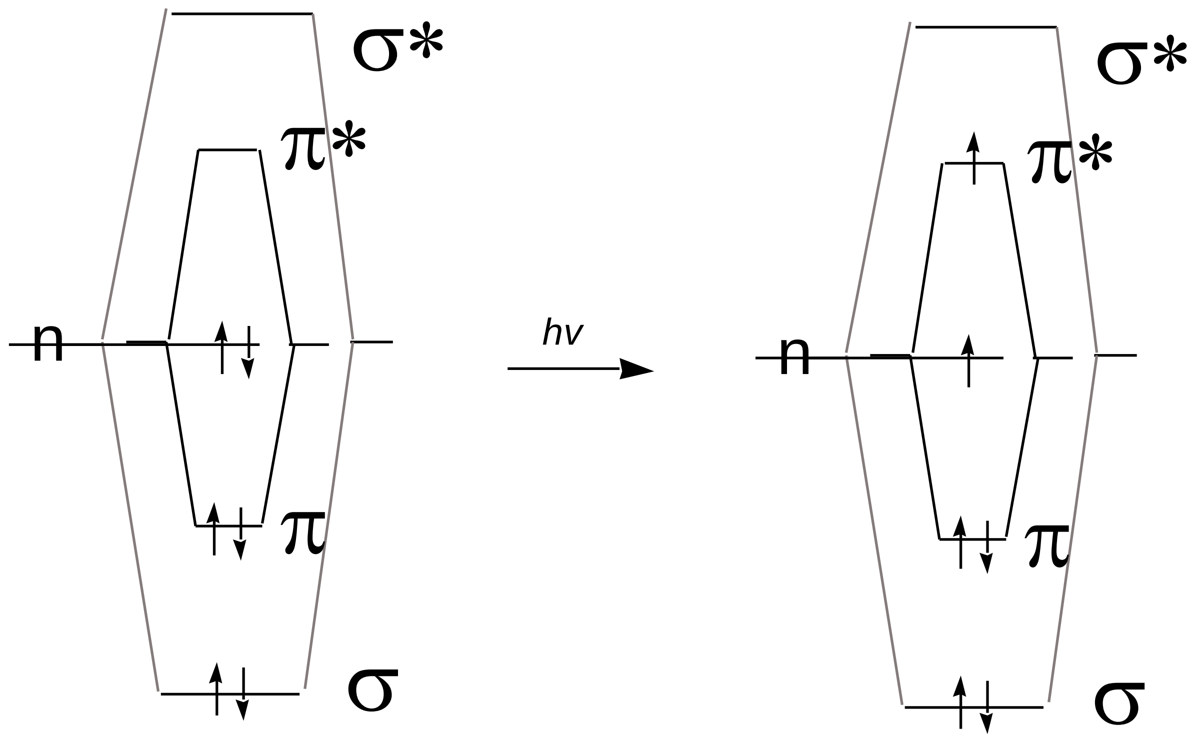


$$\pi \longrightarrow \pi^*$$

Example: ethylene absorbs at longer wavelengths:

λ_{max} = 165 nm ε= 10,000



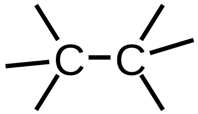
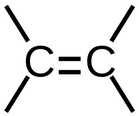
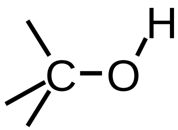
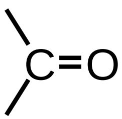


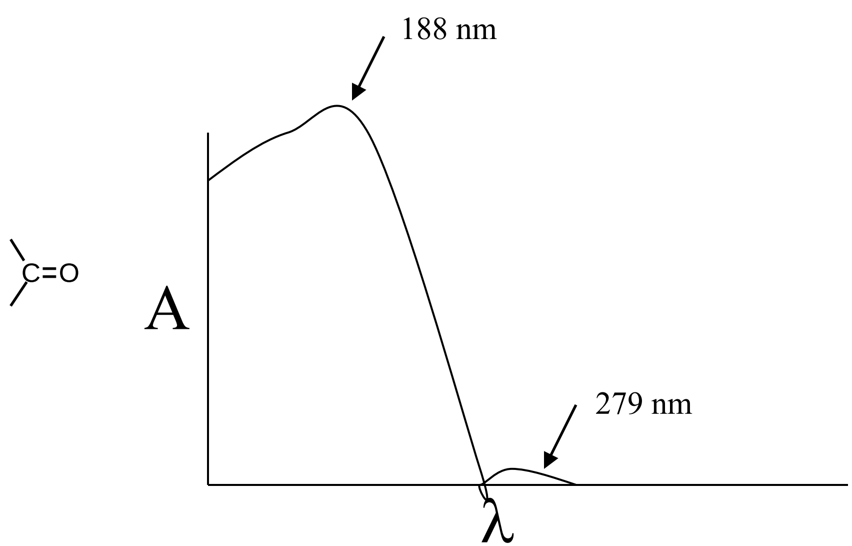
$$n \longrightarrow \pi^*$$

The n to pi* transition is at even lower wavelengths but is not as strong as pi to pi* transitions. It is said to be "forbidden."
Example:

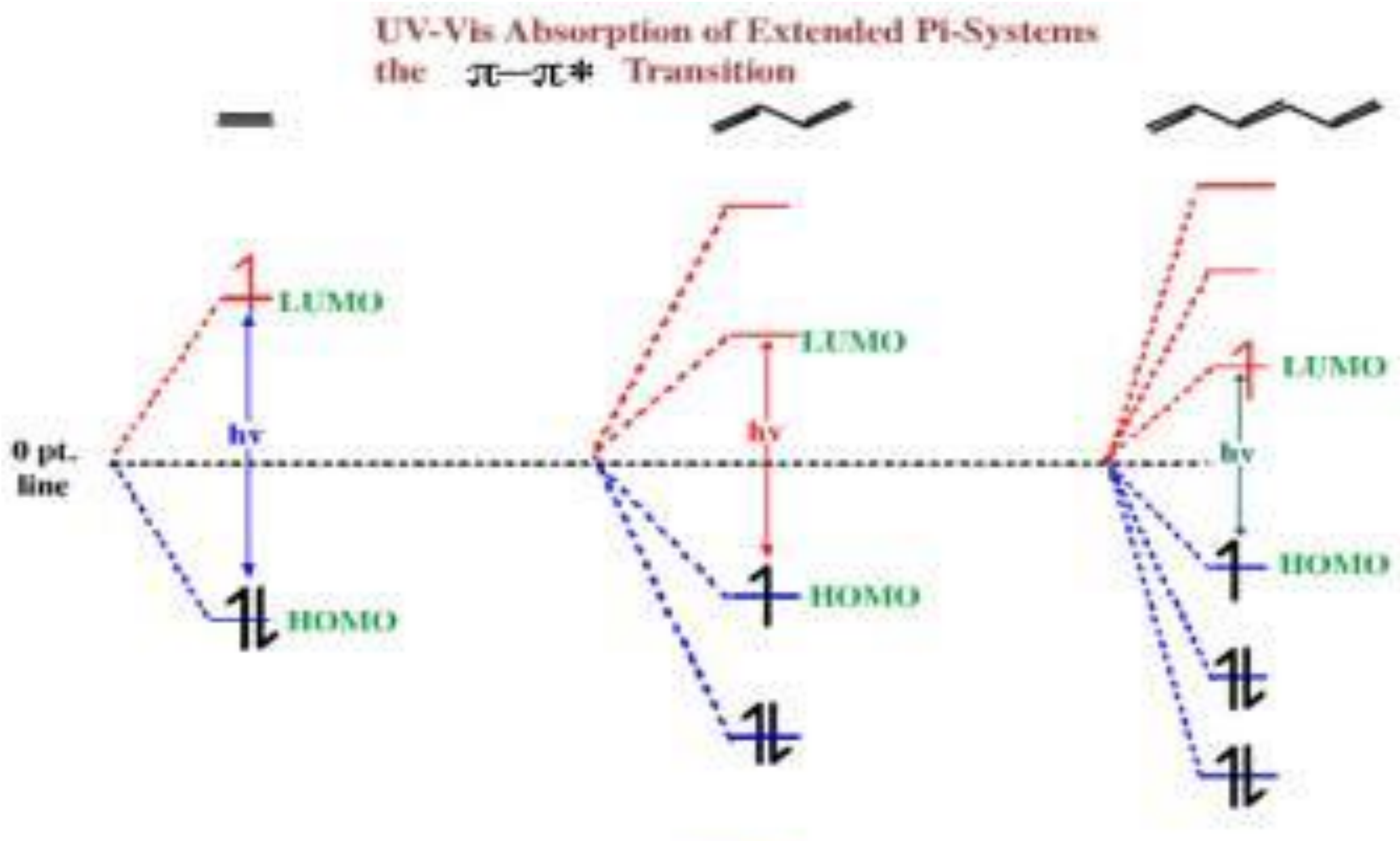
Acetone: $n{-}\sigma^*$ $\lambda_{max} = 188$ nm ; $\varepsilon = 1860$

$n{-}\pi^*$ $\lambda_{max} = 279$ nm ; $\varepsilon = 15$

	σ -> σ*	135 nm	
	π -> π*	165 nm	
	n -> σ*	183 nm	weak
	π -> π*	150 nm	
	n -> σ*	188 nm	
	n -> π*	279 nm	weak



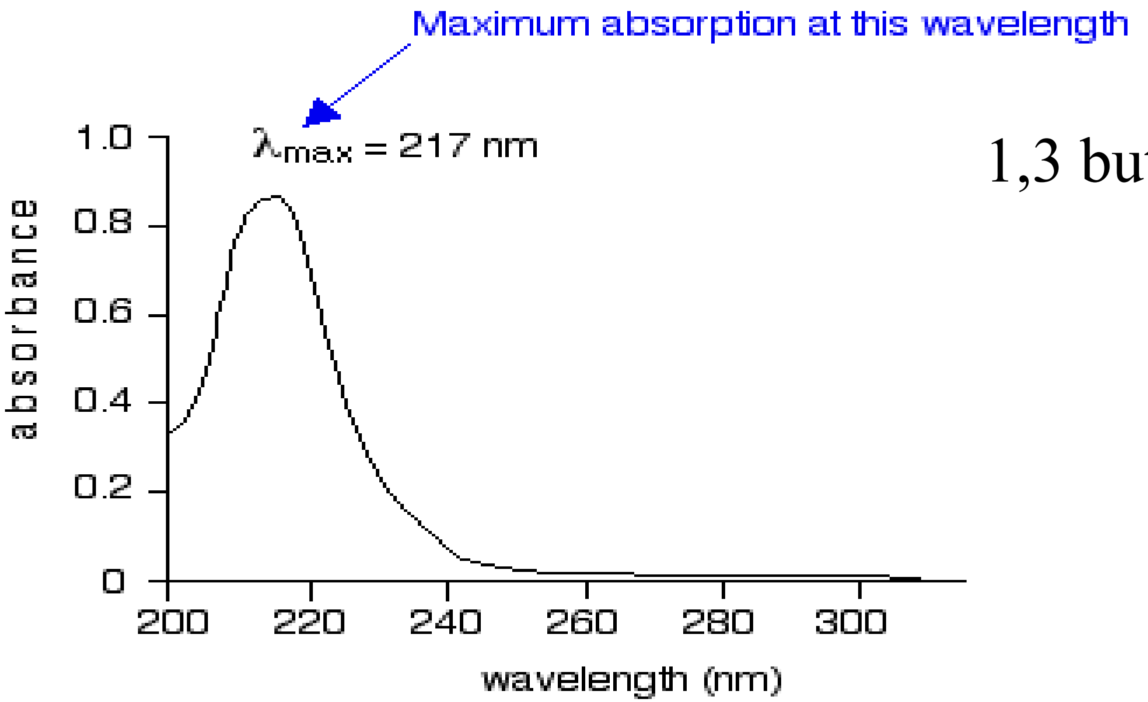
Conjugated systems:



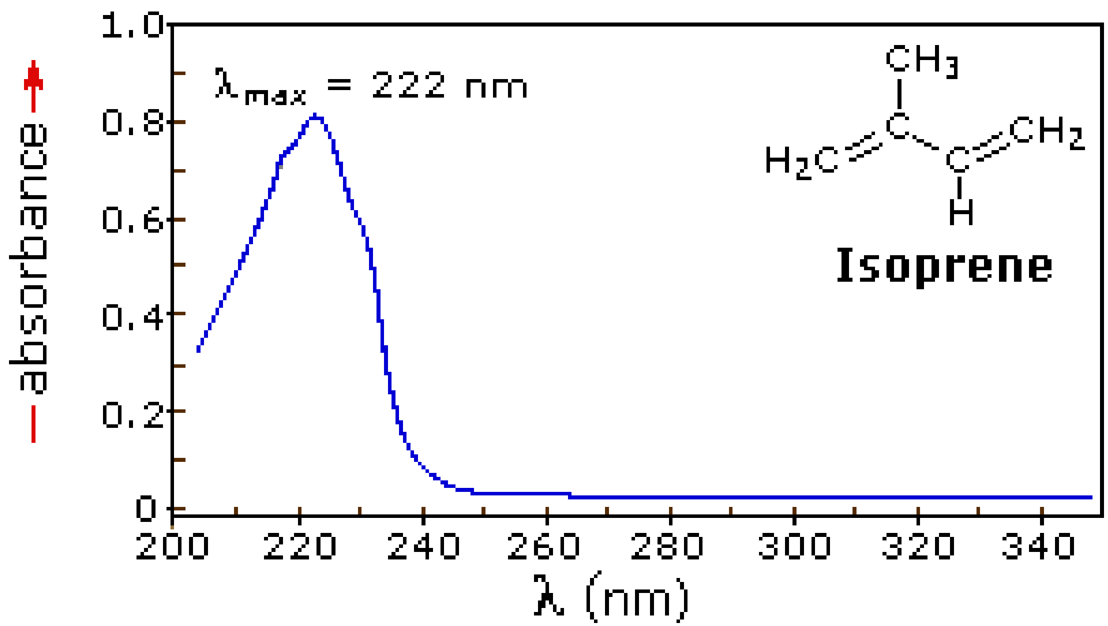
Preferred transition is between HOMO and LUMO

Note: Additional conjugation (double bonds) lowers the HOMO-LUMO energy gap:
Example:

1,3 butadiene: λ_{max} = 217 nm ; ε= 21,000
1,3,5-hexatriene λ_{max} = 258 nm ; ε= 35,000



1,3 butadiene



Typical Absorptions of Simple Isolated Chromophores

Class	Transition	λ_{max} (nm)	log ϵ	Class	Transition	λ_{max} (nm)	log ϵ
R—OH	$n \rightarrow \sigma^*$	180	2.5	$R—NO_2$	$n \rightarrow \pi^*$	271	<1.0
R—O—R	$n \rightarrow \sigma^*$	180	3.5	R—CHO	$\pi \rightarrow \pi^*$	190	2.0
$R—NH_2$	$n \rightarrow \sigma^*$	190	3.5		$n \rightarrow \pi^*$	290	1.0
R—SH	$n \rightarrow \sigma^*$	210	3.0	R_2CO	$\pi \rightarrow \pi^*$	180	3.0
$R_2C{=}CR_2$	$\pi \rightarrow \pi^*$	175	3.0		$n \rightarrow \pi^*$	280	1.5
R—C≡C—R	$\pi \rightarrow \pi^*$	170	3.0	RCOOH	$n \rightarrow \pi^*$	205	1.5
R—C≡N	$n \rightarrow \pi^*$	160	<1.0	RCOOR′	$n \rightarrow \pi^*$	205	1.5
R—N=N—R	$n \rightarrow \pi^*$	340	<1.0	$RCONH_2$	$n \rightarrow \pi^*$	210	1.5

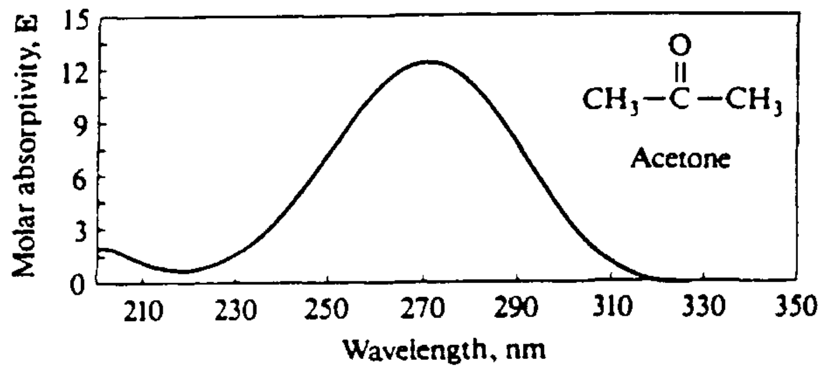
Absorption Characteristics of Some Common Organic Chromophores

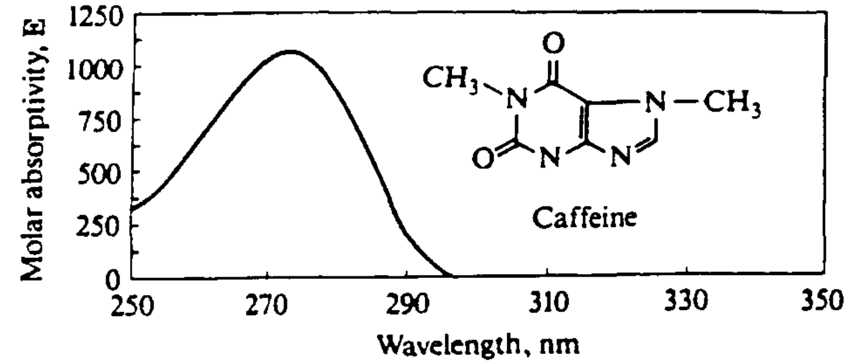
Chromophore	Example	Solvent	λ_{max}, nm	ε_{max}
Alkene	$C_6H_{13}CH{=}CH_2$	*n*-Heptane	177	13,000
Conjugate alkene	$CH_2{=}CHCH{=}CH_2$	*n*-Heptane	217	21,000
Alkyne	$C_5H_{11}C{\equiv}C{-}CH_3$	*n*-Heptane	178	10,000
			196	2,000
			225	160
Carbonyl	$CH_3\overset{\overset{\large O}{\|}}{C}CH_3$	*n*-Hexane	186	1,000
			280	16
	$CH_3\overset{\overset{\large O}{\|}}{C}H$	*n*-Hexane	180	Large
			293	12
	$CH_3\overset{\overset{\large O}{\|}}{C}NH_2$	Water	214	60
Carboxyl	CH_3COOH	Ethanol	204	41
Azo	$CH_3N{=}NCH_3$	Ethanol	339	5
Nitro	CH_3NO_2	Isooctane	280	22
Nitrate	$C_2H_5ONO_2$	Dioxane	270	12
Nitroso	C_4H_9NO	Ethyl ether	300	100
			665	20
Aromatic	Benzene	*n*-Hexane	204	7,900
			256	200

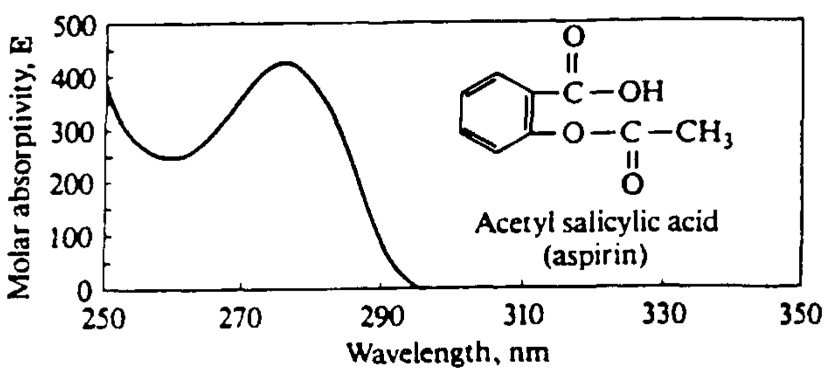
UV / visible Spectroscopy

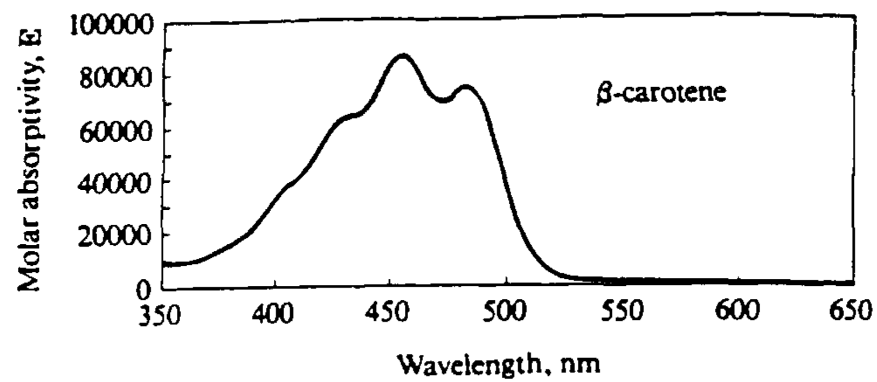
- It is often difficult to extract a great deal of information from a UV spectrum by itself.
- Generally you can only pick out conjugated systems.
- Always use in conjunction with nmr and infrared spectra.
- As structural changes occur in a chromophore it is difficult to predict exact energy and intensity changes.
- Use empirical rules.: Woodward-Fieser Rules for dienes, Woodward's Rules for enones

UV / visible Spectroscopy

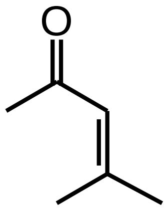




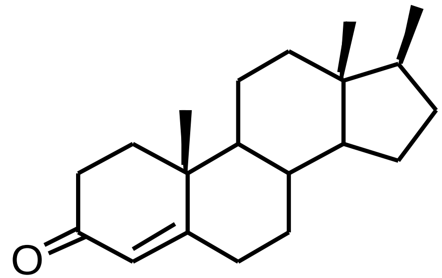




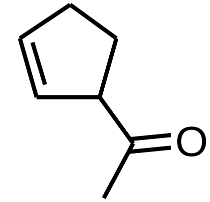
Similar structures have similar UV spectra:



λ_{max} = 238, 305 nm

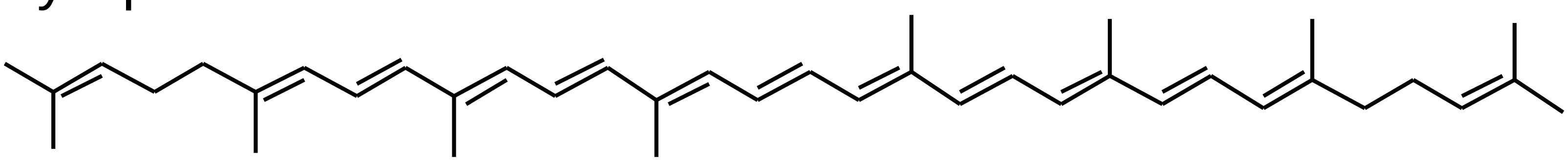


λ_{max} = 240, 311 nm



λ_{max} = 173, 192 nm

Lycopene:



λ_{max} = 114 + 5(8) + 11*(48.0-1.7*11) = 476 nm

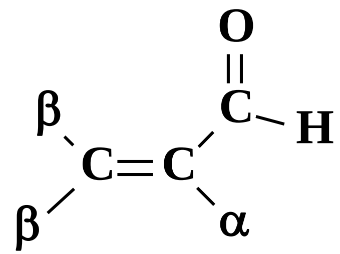
λ_{max}(Actual) = 474.

Woodward-Fieser Rules
for Predicting λ_{max}

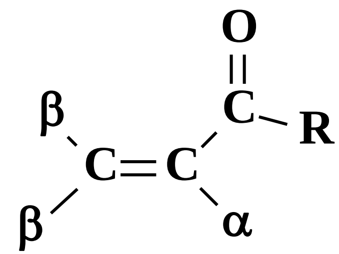
■The λ_{max} of the $\pi \rightarrow \pi^*$ transition for compounds with $\leq$ 4 conjugated double bonds can be calculated using **Woodward-Fieser rules**.

$CH_2{=}CH-CH{=}CH_2$

λ_{max} = **217 nm**



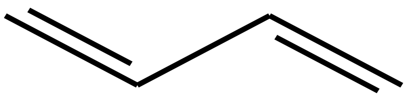
λ_{max} = **210 nm**



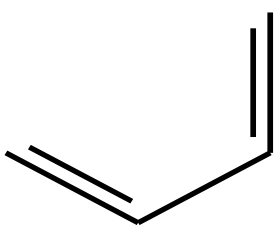
λ_{max} = **215 nm**

General base value = 215

Base value:



transoid diene 214 nm

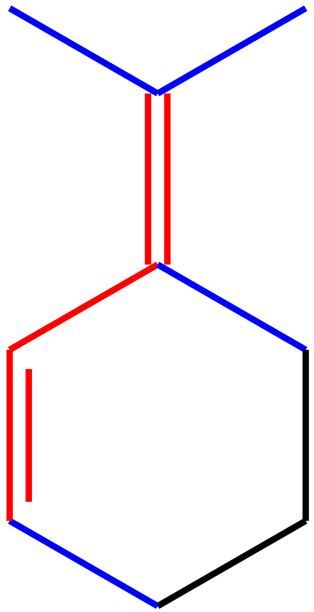


cisoid diene 253 nm

214 + 39 Homoannular system

Double bond extending conjugation	30 nm
Exocyclic double bond	5 nm
Alkyl group	5 nm
Cl, Br	5 nm
OH, OR	6 nm
SH, SR	30 nm
NH_2, NHR, NR_2	60 nm

Determination of λ_{max}



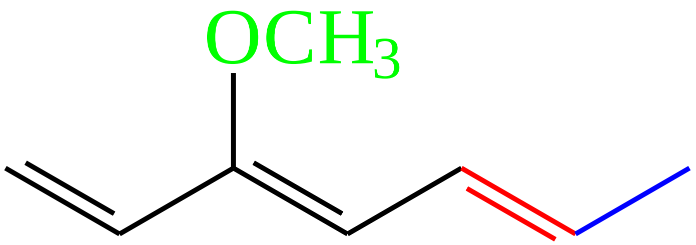
Base: 214 nm

R groups: 4 x 5 = 20 nm

exocyclic db = 5 nm

predicted λ_{max} = **239 nm**

Determination of λ_{max}



Base: 214 nm

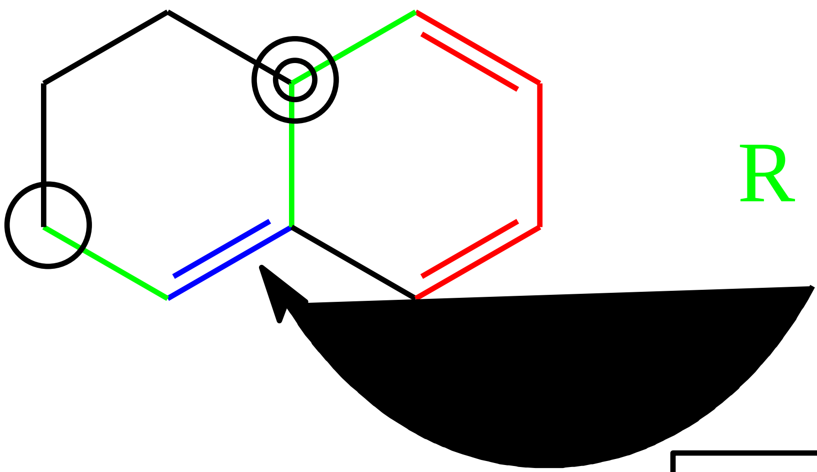
double bond ext. conj. = 30 nm

OR group = 6 nm

R group = 5 nm

predicted λ_{max} = **255 nm**

Determination of λ_{max}



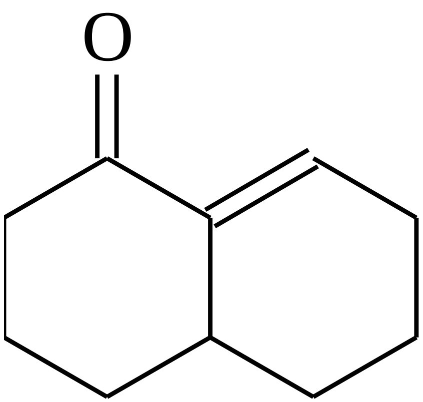
Base: 253 nm

db ext. conj. = 30

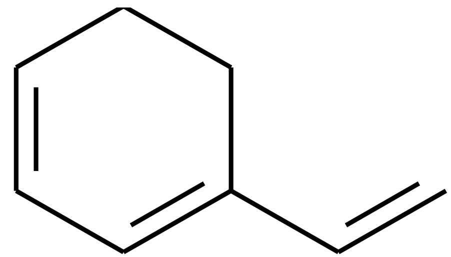
R groups: 3 x 5 = 15

exocyclic db = 5

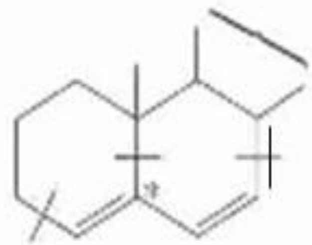
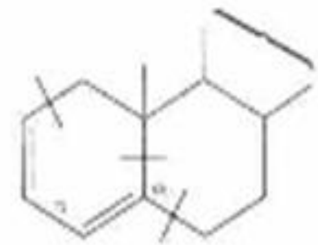
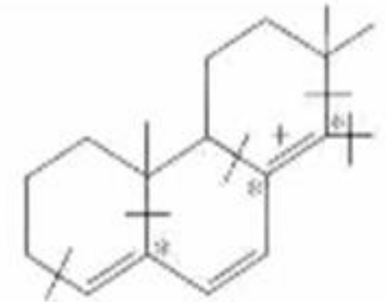
predicted λ_{max} =**303 nm**



Base	= 215
α substituent	= 10
β substituent	= 12
Exocyclic db bd	= 5
Calculated	**= 242**
Observed	= 241



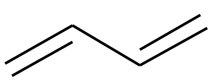
Base	= 253
Additional db bd	= 30
2 alkyl substituents	= 10
Calculated	**= 293**
Observed	= 293

			
Parent	214	214	214
alkyl substituent (5)	×3 = 15	×3 = 15	×5 = 25
exocyclic (5)	×1 = 5	×1 = 5	×3 = 15
extra conjugation (30)	×0	×0	×1 = 30
homoannular (39)	×0	×1 = 39	×0
Predicted λ_{max}(nm)	234	273	284
Observed* λ_{max}(nm)	235	275	283

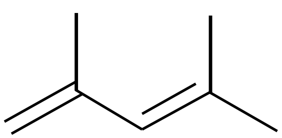
λ_{max} (nm)

Conjugation lowers the energy differences between the HOMO and LUMO energy levels, and so conjugated dienes absorb at *longer wavelengths* than isolated dienes, and trienes absorb at *longer wavelengths* than dienes, etc.

Alkyl group substitution on double bonds also causes absorption to occur at longer wavelengths.

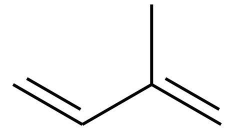


λ_{max} = 217nm



λ_{max} = 232nm

217



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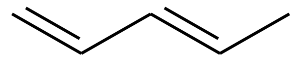

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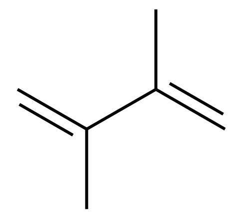

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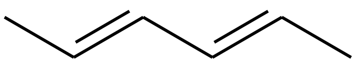

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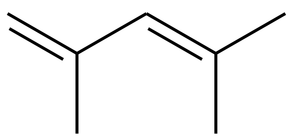

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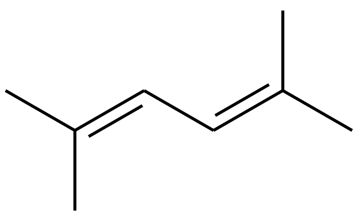


240

Quantitative Spectroscopy

- Beer Lambert Law

$$A = \log \left(\frac{I_0}{I}\right) \varepsilon \ell c$$

A = Absorbance

I_0 = intensity (power per unit area) of the incident radiation

I = intensity (power per unit area) of the transmitted radiation

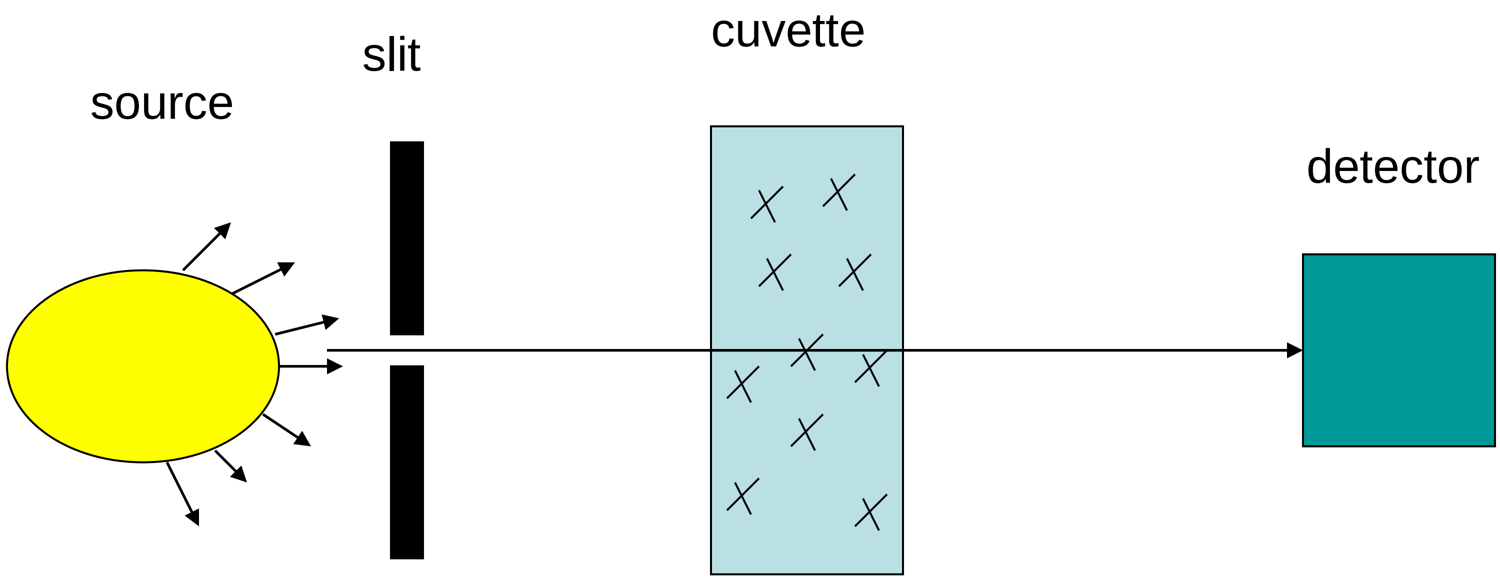
ϵ = molar absorptivity (unique for a given compound)

l = path length

c = concentration

A = - log T

T = Transmittance



- $A = -\log T = \log(P_0/P) = ebc$
- $T = P_{solution}/P_{solvent} = P/P_0$
- Works for monochromatic light
- Compound x has a unique e at different wavelengths

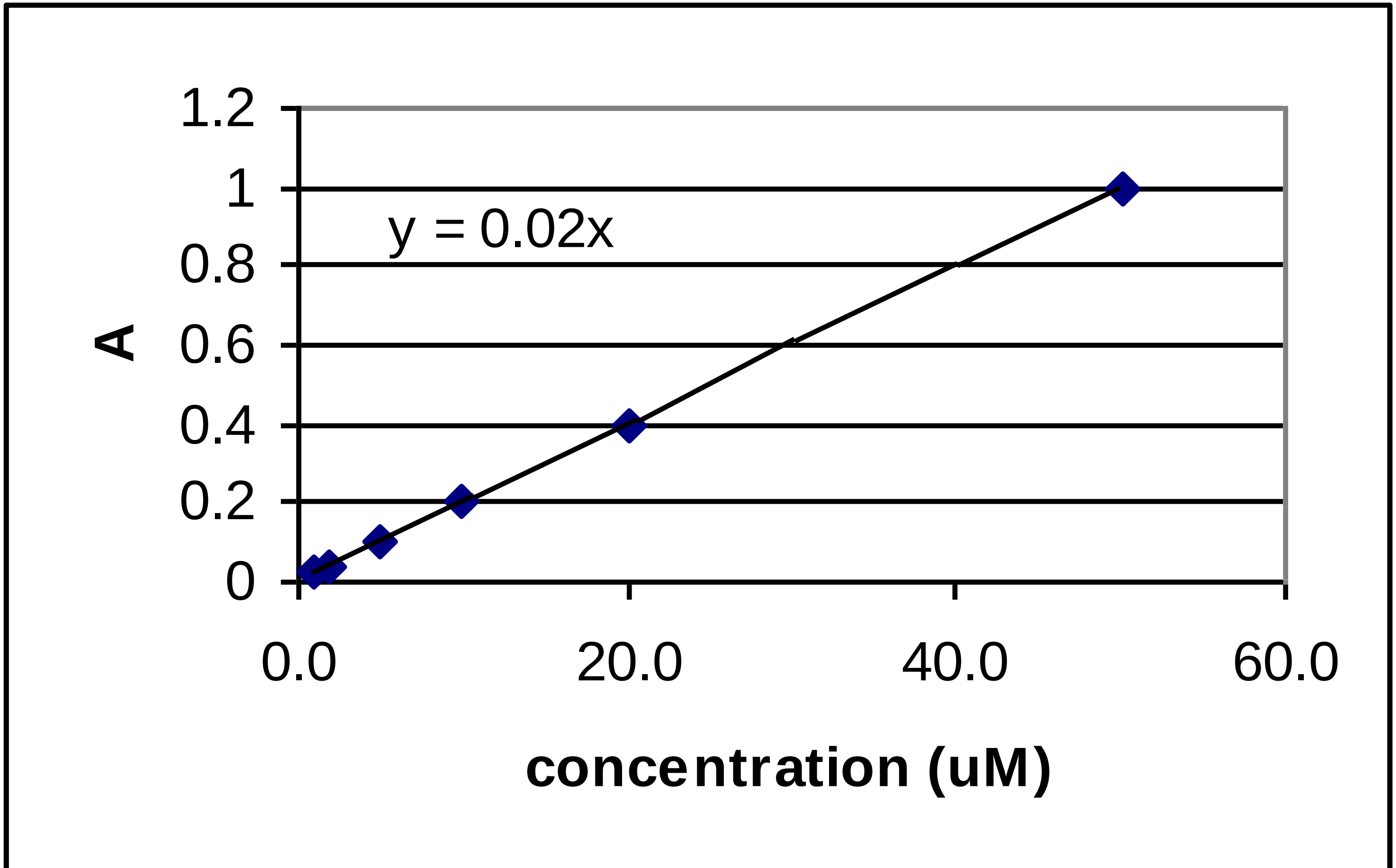
Characteristics of Beer's Law Plots

- One wavelength
- Good plots have a range of absorbances from 0.010 to 1.000
- Absorbances over 1.000 are not that valid and should be avoided
- 2 orders of magnitude

Standard Practice

- Prepare standards of known concentration
- Measure absorbance at λmax
- Plot A vs. concentration
- Obtain slope
- Use slope (and intercept) to determine the concentration of the analyte in the unknown

Typical Beer's Law Plot

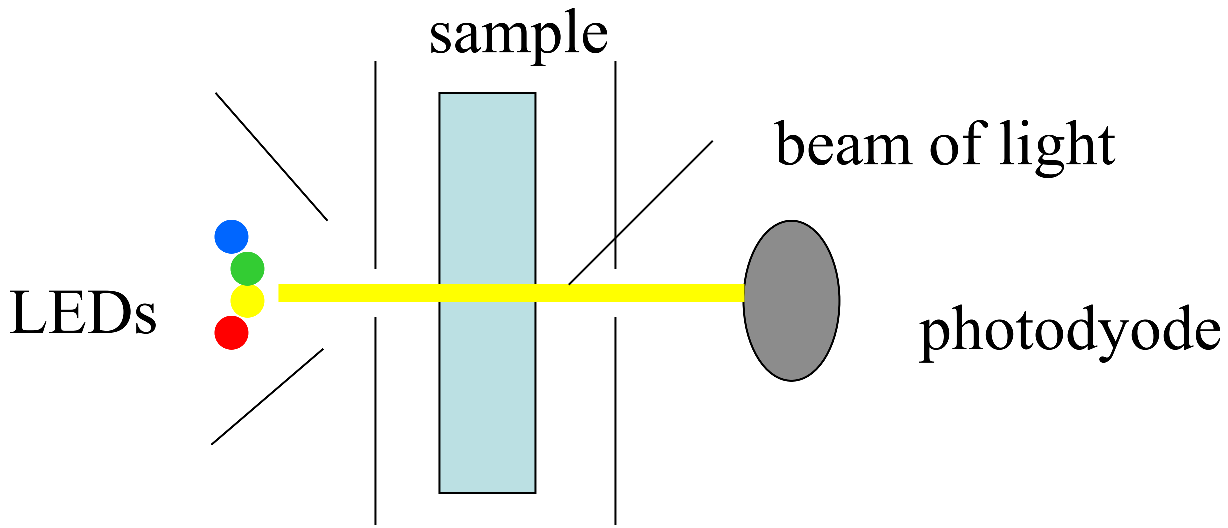


Instrumentation

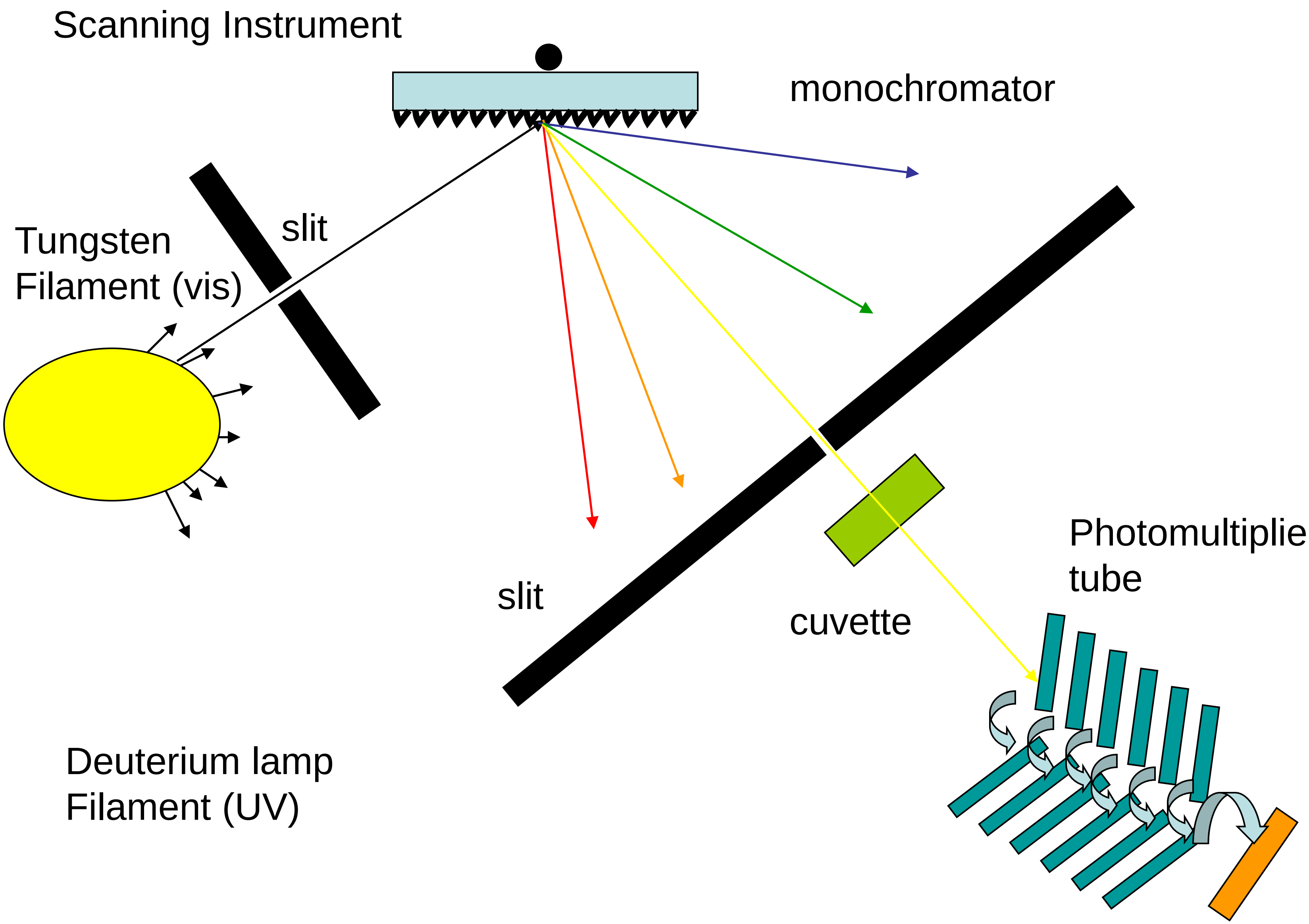
- Fixed wavelength instruments
- Scanning instruments
- Diode Array Instruments

Fixed Wavelength Instrument

- LED serve as source
- Pseudo-monochromatic light source
- No monochrometer necessary/ wavelength selection occurs by turning on the appropriate LED
- 4 LEDs to choose from



Scanning Instrument



sources

- Tungten lamp (350-2500 nm)
- Deuterium (200-400 nm)
- Xenon Arc lamps (200-1000 nm)

Monochromator

- Braggs law, nl = d(sin i + sin r)
- Angular dispersion, $dr/d\lambda = n / d(\cos r)$
- Resolution, $R = \lambda/\Delta\lambda = nN$, resolution is extended by concave mirrors to refocus the divergent beam at the exit slit

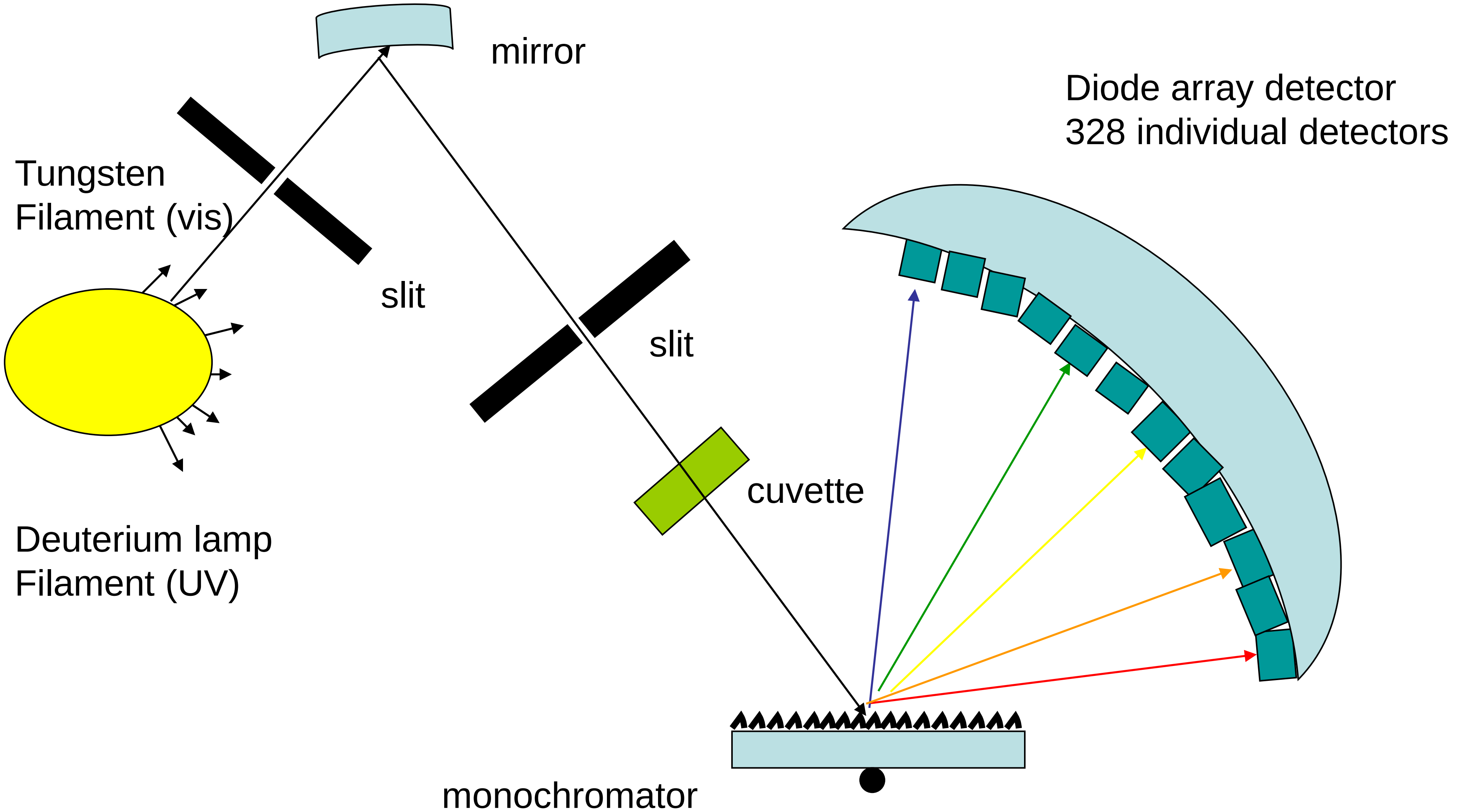
Sample holder

- Visible; can be plastic or glass
- UV; you must use quartz

Single beam vs. double beam

- Source flicker

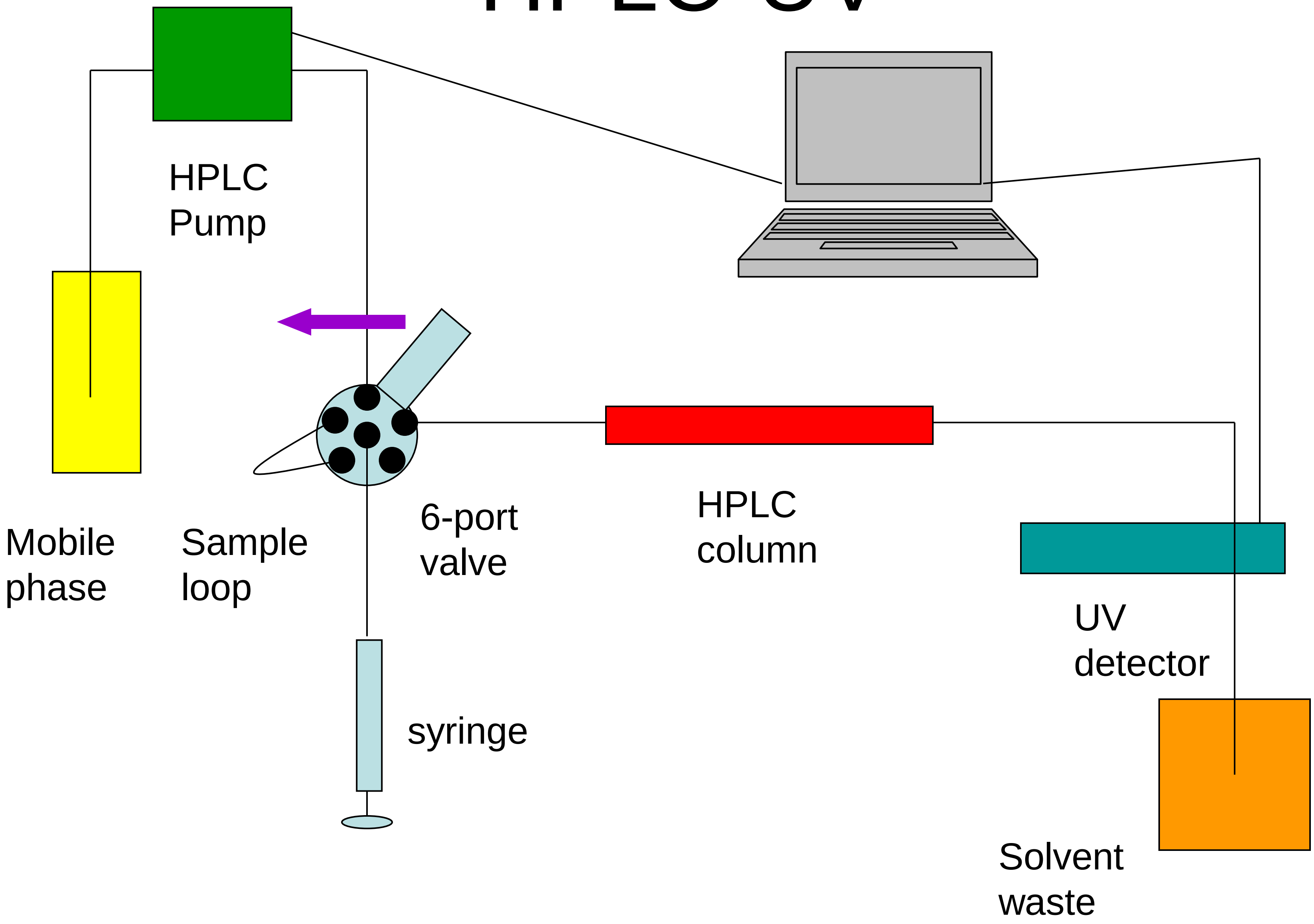
Diode array Instrument



Advantages/disadvantages

- Scanning instrument
  - High spectral resolution (63000), $\lambda/\Delta\lambda$
  - Long data acquisition time (several minutes)
  - Low throughput
- Diode array
  - Fast acquisition time (a couple of seconds), compatible with on-line separations
  - High throughput (no slits)
  - Low resolution (2 nm)

HPLC-UV


HPLC
Pump
Mobile
phase
Sample
loop
6-port
valve
HPLC
column
UV
detector
syringe
Solvent
waste